

Part I — Exact belief-state computation at scale

Exactness is not the limitation: belief-state computation at scale and the audits that bind the obstruction calculus to its worked systems

Amin Abaee

Independent Researcher

ORCID: 0000-0002-0019-1842

amin_abaee@ut.ac.ir

October 2, 2026

Abstract

This paper pushes the exact discipline of belief-state safety values into the curse-of-dimensionality regime: a four-parameter cube with $2^4 = 16$ hidden-parameter cells on 16 stock levels — 256 augmented cells, 17 controls, every computation in exact integer and rational arithmetic. The structure is classified completely. A pairwise-Hamming lemma (every pair of distinct actions sums, over any two parameter cells, to at most twice their agreement count) forces every blind survivable set to consist of Hamming-adjacent cells, and no three cells of the four-cube are pairwise adjacent: no blind policy of any period keeps three or more cells safe. The bands are exact: 16 maximal singletons at the floor's edge, exactly the 32 Hamming-adjacent pairs at every level above it (the matched-singly-mismatched alternation rises at net $\frac{1}{5}$ per two steps with a $\frac{1}{10}$ dip). The observation ladder doubles: each additional exact probe of one parameter doubles the conditional kernel mass — $\frac{1}{16} \rightarrow \frac{1}{8} \rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1$ at the edge, $\frac{1}{8} \rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1 \rightarrow 1$ above it — so three probes exhaust observation above the edge. Exact point-based evaluation is verified against definition-level enumeration at seven rational beliefs, three levels, four horizons; the four-step alpha-set deduplicates 83,521 action sequences to at most 857 distinct outcome vectors on the tested levels (545 at $z_0 = 2.5$). The antichain census is the headline: 1,048,576 raw subset evaluations collapse to 496 stored maximal sets. The instance is then enlarged to five parameters (Section 0.9), where the classification changes qualitatively and is given completely: the maximal blind-survivable sets are the 32 radius-one Hamming balls at every level, plus 80 four-cell sets appearing from $z_0 \geq 1.3$, so that $16 \times 2^{32} = 68,719,476,736$ raw subsets collapse to 1,552 stored sets. The raw lattice grows $65,536\times$ against a $3.13\times$ growth in the stored object, improving compression by four orders of magnitude; and the pairwise-Hamming instrument, sharp at four parameters, is shown necessary but no longer sufficient at five. A third instance of the theory's additive deadline law closes the paper (threshold $1 + T/2$, zero probe excursion).

Keywords: exact computation; partially observable control; zero-suppressed sets; survivable sets; Hamming structure; observation design; antichain compression

0.1 Introduction

Two working disciplines govern the exact treatment of belief-state safety values: evaluate exactly where you probe, and store the antichain of maximal survivable sets rather than the lattice (Abaee, 2026, An obstruction calculus). This paper asks whether the exact discipline survives the curse of dimensionality, and what new structure appears there. On the instance studied here the answer is affirmative, and the structure changes: the four-parameter cube is not a larger version of the two-parameter instance but a different regime. The cube is the paper's running instance; every structural claim below is instantiated on it. Its survivable sets are governed by Hamming adjacency rather than by coordinate matching; its observation ladder is geometric rather than one-shot; and its raw subset lattice (2^{16} per level) is four orders of magnitude larger than its stored antichain. All claims below are regenerated by the paper's verification script in exact integer and rational arithmetic, standard library only. The contributions are: the pairwise-Hamming classification of the cube's survivable-set structure (Section 0.3); the exact observation-ladder doubling and the point-based evaluation at four parameters (Sections 0.4–0.6); and the antichain census with the audited-structure classification and the deadline instance (Sections 0.7–0.8).

0.2 The cube instance

The cube's local vocabulary (z , z_0 , T , the cell structure) is disjoint from the companion papers' colliding letters; the horizon T carries the same meaning as the certification companion's $J_{\mathcal{I}}(T)$. The state is (z, θ) with stock z on the grid $z_0 \in \{1.0, 1.1, \dots, 2.5\}$ and hidden parameter $\theta = (\theta_1, \theta_2, \theta_3, \theta_4) \in \{+1, -1\}^4$: 16 parameter cells on 16 levels, 256 augmented cells. Controls are $u \in \{+1, -1\}^4 \cup \{(0, 0, 0, 0)\}$ (17 actions), and the one-step drift is

$$\dot{z} = d(u, \theta) = -\frac{1}{2} + \frac{1}{5} \sum_{i=1}^4 u_i \theta_i :$$

the matched action $u = \theta$ rises at $+\frac{3}{10}$; a single mismatch gives $-\frac{1}{10}$, two give $-\frac{1}{2}$, three give $-\frac{9}{10}$, the antimatched action $-\frac{13}{10}$, and the hold action $-\frac{1}{2}$ for every cell. The floor is $z \geq 1$. The sum over all sixteen cells of any action's drifts is -8 per step: the cube as a whole always declines, and the blind controller declared here sees no observation of θ .

0.3 The pairwise-Hamming classification

Lemma 1 (pair-sum bound and its survival cost). *Fix the dimension $m \geq 1$, cells $\theta, \theta' \in \{+1, -1\}^m$ at Hamming distance h , the action set $U = \{+1, -1\}^m \cup \{(0, \dots, 0)\}$, the drift $d(u, \theta) = -\frac{1}{2} + \frac{1}{5}\langle u, \theta \rangle$, the floor $z \geq 1$, and the blind class: one action sequence for every parameter branch, of any period and possibly aperiodic. Then: (i) for every $u \in \{+1, -1\}^m$, $\langle u, \theta \rangle + \langle u, \theta' \rangle \leq 2(m - h)$, and the hold action gives 0 for every pair; (ii) if a blind policy keeps both cells above the floor indefinitely, then the pair's long-run average inner product satisfies $\liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t < T} (\langle u_t, \theta \rangle + \langle u_t, \theta' \rangle) \geq 5$.*

Proof. (i) The coordinates split into $m - h$ agreements and h disagreements; on an agreement $u_i \theta_i = u_i \theta'_i \leq 1$ and on a disagreement the two terms cancel, so the sum is $2 \sum_{\text{agree}} u_i \theta_i \leq 2(m - h)$; the hold action gives 0 termwise. (ii) Survival of cell θ means $z_0 + \sum_{t < T} d(u_t, \theta) \geq 1$ for every T ; dividing by T and passing to the limit, the Cesàro average of the drifts has $\liminf \geq 0$, and since $d(u_t, \theta) = -\frac{1}{2} + \frac{1}{5}\langle u_t, \theta \rangle$, the $\liminf$ of the average of $\langle u_t, \theta \rangle$ is at least $\frac{5}{2}$; the same holds for θ' , and the two $\liminf$ s add. For a periodic policy the average is the exact cycle average and no $\liminf$ is needed. $\square$

Corollary 2 (the cube's pairs are Hamming-adjacent). *At $m = 4$: (ii) forces the pair average to reach 5 while (i) caps it at $2(4 - h)$, so $5 \leq 2(4 - h)$, i.e. $h \leq \frac{3}{2}$: every blind survivable pair has $h \leq 1$, and since survivability passes to subsets, every pair inside a blind survivable set is Hamming-adjacent. In general the argument gives $h \leq m - \frac{5}{2}$ (Remark 5).*

Lemma 3 (no adjacency triangles). *No three distinct cells of $\{+1, -1\}^m$ (any m) are pairwise at Hamming distance 1. Hence every blind survivable set at $m = 4$ has at most two members, under every blind policy — any period, and no period at all. The bound (i) of Lemma 1 is verified for all 120 cell pairs and all 17 actions at $m = 4$ (and for all 496 pairs and 33 actions at $m = 5$), and Lemma 3 over all $\binom{16}{3} = 560$ triples, by the verification script.*

Proof. Flip coordinates so that the first cell is **1** (flips preserve all Hamming distances). A second cell at distance 1 is **1** with exactly one coordinate i flipped. A third distinct cell at distance 1 from **1** is **1** with exactly one coordinate j flipped: if $j = i$ it is the second cell, and if $j \neq i$ the two differ from each other in the two coordinates i, j , so their Hamming distance is 2. No triangle exists. (Equivalently: the hypercube is bipartite by coordinate parity, so it has no odd cycle.) $\square$

Proposition 4 (bands and maximality). *A singleton $\{\theta\}$ is survivable from every $z_0 \geq 1$ (matched play rises at $+\frac{3}{10}$). A Hamming-adjacent pair $\{\theta, \theta'\}$ is survivable exactly from $z_0 \geq 1.1$: the alternation (θ, θ') gives each member drifts $+\frac{3}{10}, -\frac{1}{10}$ per two steps — net $+\frac{1}{5}$ with a within-cycle dip of $\frac{1}{10}$ — certified at 60 steps, and from $z_0 = 1.0$ the first cell to act under its mismatched action dips below the floor. The maximal survivable sets are therefore exactly the 16 singletons at $z_0 = 1.0$ and exactly the 32 Hamming-adjacent pairs at every $z_0 \geq 1.1$, and nothing larger under any blind policy. An exhaustive search over all 5,219 one-, two-, and three-periodic policies with 60-step certificates reproduces this classification independently.*

Proof. Singletons: matched play has drift $+\frac{3}{10}$ from every $z_0 \geq 1$. Pairs: let $\{\theta, \theta'\}$ be Hamming-adjacent and alternate (θ, θ') ; each member takes the matched drift $+\frac{3}{10}$ and the mismatched drift $-\frac{1}{10}$ once per cycle, a net of $+\frac{1}{5}$ per two steps with the cycle minimum $z_0 - \frac{1}{10}$ after the mismatched step — survivable exactly from $z_0 \geq 1.1$ (the 60-step certificates confirm the cycle arithmetic). No pair from $z_0 = 1.0$: for any first action u , if $u = \theta$ then $\langle u, \theta' \rangle = 2$ and cell θ' falls to $z_0 - \frac{1}{10} < 1$ in one step, while any other u drives at least one member below the floor in its first step (the hold gives $-\frac{1}{2}$ to both); no repair exists because the floor binds at every step. Nothing larger: any set of three or more cells contains two at Hamming distance ≥ 2 (Lemma 3), and that pair violates Corollary 2, so no blind policy keeps it above the floor — and survivability passes to subsets, so the whole set fails with it. Maximality is immediate: singletons extend to pairs from 1.1, pairs extend to nothing. $\square$

Remark 5 (scope in dimension: the ball construction). The pairs-only census is the four-dimensional cube's, and the boundary is sharp. In general dimension the pair argument of Corollary 2 gives only $h \leq m - \frac{5}{2}$; and the constant matched action $u = \mathbf{1}$ gives a cell at Hamming distance k the constant drift $-\frac{1}{2} + \frac{m-2k}{5}$, nonnegative exactly when $k \leq \frac{2m-5}{4}$. Hence the Hamming ball of radius $r^*(m) = \lfloor (2m-5)/4 \rfloor$ around any cell is blind-survivable from every $z_0 \geq 1$: $r^*(4) = 0$, $r^*(5) = r^*(6) = 1$, $r^*(7) = r^*(8) = 2$. At $m = 5$ the radius-one ball — $1+5 = 6$ cells, drifts $+\frac{1}{2}$ and $+\frac{1}{10}$ under constant matched play — survives from the floor's edge, where the four-cube keeps only singletons: the classification changes qualitatively at $m = 5$, and the maximal sets for $m \geq 5$ are classified in Section 0.9 at $m = 5$, and left open at $m \geq 6$. All eight radii and the sign pattern are verified by the verification script, with 60-step simulations of the radius-one ball at $m = 5$ and the 29-cell radius-two ball at $m = 7$.

Remark 6 (the crude instrument is not sufficient). The averaging identity used on smaller instances — a survivable set S needs $\sum_i |\sum_{\theta \in S} \theta_i| \geq \frac{5}{2} |S|$ — is not sufficient here: the four-set $\{++++, ++-+, +-+-, -++-\}$ attains the bound with equality ($4+2+2+2 = 10$) yet is not survivable, since it pairs cells at Hamming distance 2. The pairwise instrument of Lemma 1 is the sharp one.

0.4 The observation ladder doubles

Proposition 7 (geometric ladder). *Prior b_0 is uniform on the sixteen cells; a probe of one parameter refines the support to a subcube. The maximal conditional kernel mass doubles with each probe:*

$$\begin{aligned} \frac{1}{16} &\rightarrow \frac{1}{8} \rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1 & (z_0 = 1.0), \\ \frac{1}{8} &\rightarrow \frac{1}{4} \rightarrow \frac{1}{2} \rightarrow 1 \rightarrow 1 & (z_0 \geq 1.1), \end{aligned}$$

for zero, one, two, three, and four probes. Above the band's edge three probes exhaust observation — the support is then a Hamming-adjacent pair, jointly survivable, and the fourth probe buys nothing — while at the edge the fourth probe is worth $\frac{1}{2}$, the single-cell mass the pair's dip excludes.

Proof. The companion theory's antichain formula makes the kernel value of a prior the maximal survivable-set

mass it charges, and survivability is hereditary downward — a policy keeping a set alive keeps every subset alive — so the value at the uniform prior on a support S is $\max_{A \subseteq S, A \text{ survivable}} |A|/|S|$, attained inside the support. By Proposition 4 the survivable subsets of a subcube are singletons and Hamming-adjacent pairs, so a 2^j -cell subcube has value $2/2^j$ from $z_0 \geq 1.1$ and $1/2^j$ at the edge. The supports after 0, 1, 2, 3, 4 probes of one parameter are subcubes with $j = 4, 3, 2, 1, 0$, giving the two ladders exactly; the fourth edge probe buys the last cell because at the edge only singletons survive and the fully probed support is then itself the surviving cell. The script recomputes all ten values by exhaustive survivable-subset enumeration within each support. $\square$

0.5 Exact point-based evaluation at four parameters

The evaluation is point-based in the POMDP literature’s sense (Lovejoy, 1991), with the distinction that the values here are exact at the probed beliefs, not bounds.

Proposition 8 (exact alpha-set values). *The alpha-vector representation of the underlying theory (its class-restricted rational alpha-vectors) evaluated at probed beliefs returns the exact value: for horizons $k = 1, 2, 3, 4$, three stock levels, and seven random rational beliefs, the maximum over the deduplicated alpha-set equals the maximum over definition-level trajectory enumeration — all 84 pairings equal. The four-step alpha-set deduplicates $17^4 = 83,521$ action sequences to at most 857 distinct outcome vectors on the tested levels (545 at the top level) (and to a single vector at the top level’s one-step horizon, where every action is safe — the silent region of the cube). Point-based selection chooses where to evaluate, never what the value is.*

Proof. Both sides are maxima of the same linear functional over the same finite set of action sequences: the alpha-set is the image of the sequence set under the map that records, for each parameter cell, whether the sequence survives at the level; the maximum of a function over a finite set equals the maximum over its image, and deduplication removes only repeated vectors, never the attaining one. Exactness of each entry is the definition-level simulation; no discretization enters. $\square$

0.6 The antichain census

Proposition 9 (the antichain, not the lattice). *The raw subset lattice carries $2^{16} = 65,536$ subsets per level — $16 \times 65,536 = 1,048,576$ over the grid — while the stored object is the antichain of maximal survivable sets — the zero-suppressed discipline of representing combinatorial set families by their maximal elements (Minato, 1993): 16 sets at the edge and 32 per level above it, 496 stored sets in total. The per-level compression ratio is $4,096 \times$ at the edge and $2,048 \times$ above it; and by Lemma 3 nothing is lost — the antichain is the whole classification, not a summarization of one.*

Proof. The raw count is $16 \times (2^{16} - 1) = 1,048,560$ nonempty subsets; the headline uses the full lattice $16 \times 2^{16} = 1,048,576$, the larger and therefore conservative figure. The stored count is $16 + 15 \times 32 = 496$ by Proposition 4. The antichain is lossless by the theory’s kernel

Table 1: The cube’s audited structure, one row per layer; every entry is a verification-script output in exact rational arithmetic.

Layer	Exact statement
Per-step drift	matched $+\frac{3}{10}$; k mismatches: $+\frac{3}{10} - \frac{2k}{5}$; hold $-\frac{1}{2}$
Cube accounting	drifts sum to -8 per step over the 16 cells
Singletons	16 survivable (exactly the floor’s edge)
Pairs	exactly the 32 Hamming-adjacent pairs; no other pair
Larger sets	none (three-set and four-set candidates fail)
Raw vs stored	$2^{16} = 65,536$ raw subsets per level, 496 stored maximal sets
Observation ladder	ten probe schedules, values doubled at each rung
Deadline law	$z_0 \geq 1 + T/2$, $T = 0, \dots, 4$ (all cells)

formula: the value of any prior is the maximal survivable-set mass, read off the stored sets alone. $\square$

0.7 The audited structure at a glance

0.8 A deadline instance

Proposition 10 (the deadline law on the cube). *Hold for T steps (drift $-\frac{1}{2}$), let the parameter be revealed exactly at T , then play matched forever (drift $+\frac{3}{10}$ on every branch). The companion theory’s additive deadline law applies with blind-drift cost $d_0 = \frac{1}{2}$ and zero probe excursion: viability holds exactly when*

$$z_0 \geq 1 + \frac{T}{2}, \quad T = 0, \dots, 4,$$

verified by direct simulation on the grid and, independently, by a mechanized proof (Section 0.14). The excursion vanishes because the learned act rises on every branch immediately — the law’s e_p term prices the probe step the free revelation does not need.

Proof. Necessity: under the hold every cell drifts $-\frac{1}{2}$, so at revelation every branch sits at $z_0 - T/2$, and survival to revelation forces $z_0 - T/2 \geq 1$. Sufficiency: the hold keeps every branch at $z_0 - t/2 \geq 1$ for $t \leq T$ exactly under the same inequality, and after revelation the matched action rises at $+\frac{3}{10}$ on every branch, so no later constraint binds. This is the theory’s additive deadline law at $d_0 = \frac{1}{2}$, $e_p = 0$; the direct branchwise argument needs no scalar-additivity hypothesis, and the simulation certifies the identity cell by cell on the grid. The mechanized layer proves the same threshold as an *equivalence* — viability holds iff $z_0 \geq 1 + T/2$ — over an arbitrary ordered field and with T symbolic, so neither the grid nor the restriction $T \leq 4$ is load-bearing for the argument. $\square$

Reciprocal check (computational-certification companion). The deadline law’s radius structure is certified externally by the certification companion (Abaee, 2026, Computational viability certification): on this instance’s audited domain, a radius- δ belief cell inherits the verdict exactly for $\delta \leq z_0 - (1 + T/2)$, with $\delta = 0$ recovering the stored per-state bound and the boundary cells zero-slack; the joint check is archived with the companion’s verification record.

0.9 Beyond four parameters: the five-cube, completely classified

Remark 5 fixes the ball threshold and records that the classification changes qualitatively at $m = 5$. This section carries the enlargement through: it gives every maximal blind-survivable set of the five-cube, at every level, together with the census and deduplication figures. Every claim below is settled by the safety game of Section 0.14 — a greatest-fixed-point computation on the capped non-negative orthant, in exact integer arithmetic — and not by an averaging bound. That distinction is load-bearing, and Remark 6 is only the first place it bites.

0.9.1 The five-cube instance

Take $m = 5$: $2^5 = 32$ hidden-parameter cells over the same sixteen stock levels, so 512 augmented cells against the four-cube's 256, and $2^5 + 1 = 33$ actions against 17. The drift law and the floor are unchanged, so from $\langle u, \theta \rangle = m - 2 \text{dist}(u, \theta)$ the drift in tenths is

$$D(h) = -5 + 2m - 4h = 5 - 4h \quad (m = 5),$$

namely $+\frac{1}{2}$ at distance 0, $+\frac{1}{10}$ at distance 1, then $-\frac{3}{10}, -\frac{7}{10}, -\frac{11}{10}, -\frac{3}{2}$ at distances 2, 3, 4, 5, with $-\frac{1}{2}$ for the hold. Two things change at once: the matched cell rises faster ($+\frac{1}{2}$ against $+\frac{3}{10}$), and — the qualitative change — a cell at Hamming distance 1 now *rises*, where at $m = 4$ it fell. That is the statement $r^*(4) = 0$, $r^*(5) = 1$ of Remark 5 read at the drift table.

0.9.2 The maximal sets

Proposition 11 (the five-cube's maximal blind-survivable sets). *For the five-cube instance above, the maximal blind-survivable sets are:*

1. the 32 Hamming balls of radius 1, each of $1 + 5 = 6$ cells, at every level $z_0 \geq 1.0$;
2. from $z_0 \in \{1.0, 1.1, 1.2\}$, nothing else — 32 maximal sets per level;
3. from $z_0 \geq 1.3$, additionally one orbit of four-cell sets under the 3,840-element automorphism group (XOR by any cell, then any permutation of the five coordinates), of size 80, with pairwise-distance multiset $\{1, 1, 1, 1, 2, 2\}$ — $32 + 80 = 112$ maximal sets per level.

No set of five or more cells other than the balls is maximal, and no set of three or fewer is maximal.

Proof. Candidates. By Corollary 2 generalised, survival of a pair needs $h \leq m - \frac{5}{2}$, i.e. $h \leq 2$ at $m = 5$, so every survivable set is a clique of the graph on the 32 cells with edges at distance ≤ 2 , and every such set lies in a maximal clique. That graph has 192 maximal cliques: 32 of size 6 and 160 of size 4.

Size 6. The 32 size-6 maximal cliques are exactly the radius-1 balls. Under the constant matched action $u = v$ on the ball $B(v, 1)$, the centre receives $D(0) = +5$ and each of the five neighbours $D(1) = +1$, all strictly positive, so the slack vector grows from $(L, \dots, L)$ at every level, $L = 0$ included. All 32 are therefore survivable from the floor's edge.

Size 4. The 160 size-4 maximal cliques fall into exactly two orbits under the automorphism group, 80 each: the *tetrahedral* orbit, all six pairwise distances equal to 2, and the orbit with distance multiset $\{1, 1, 1, 1, 2, 2\}$. The tetrahedral orbit is *not* survivable at any finite starting stock: choose its representative 00000, 11000, 10100, 01100. Across these four cells, the first three signed coordinates cancel and the remaining two contribute at most 8 to the aggregate; in integer-tenths units each sign action has aggregate drift at most $-20 + 2(8) = -4$, and hold is still worse. Their total slack therefore decreases indefinitely under any common action sequence. The other orbit is the two-coordinate square face with representative 00000, 10000, 01000, 11000. Every sign action has distance at least two from one square vertex, giving its first-step drift at most -3 tenths; hold drifts -5 . Thus $L < 3$ cannot survive its first step. At $L \geq 3$, alternate the actions matching opposite corners 00000 and 11000. Every vertex gains 2 tenths per two steps, with initial dip at most 3; hence it survives indefinitely. Orbit symmetry transfers both arguments to all $80 + 80$ four-cliques, with no artificial state cap.

Smaller sets. All 912 cliques of size ≤ 3 are contained in some radius-1 ball, and survivability is hereditary downward, so none is maximal. A size-5 clique would lie in a maximal clique of size ≥ 5 , hence in a ball, hence is non-maximal. $\square$

Remark 12 (the pairwise instrument is necessary but no longer sufficient). At $m = 4$ the pairwise-Hamming bound is sharp: it excludes every pair at distance ≥ 2 , and everything it permits is survivable. At $m = 5$ it is *not* sharp. The tetrahedral orbit has all six pairwise distances equal to 2, which meets the bound $h \leq 2$ with equality, and no blind policy of any period keeps it above the floor. So the pair-sum condition of Lemma 1 is necessary and not sufficient from dimension five onward, and the classification above cannot be recovered from pairwise data alone. Remark 6 records the same failure for the cruder averaging instrument at $m = 4$; the point here is that the sharper instrument inherits the failure one dimension later.

0.9.3 The census at five parameters

The raw subset lattice at $m = 5$ carries $2^{32} = 4,294,967,296$ subsets per level — $16 \times 2^{32} = 68,719,476,736$ over the grid — against the four-cube's 1,048,576. By Proposition 11 the stored antichain is

$$3 \times 32 + 13 \times 112 = 96 + 1,456 = 1,552$$

maximal sets. Per level the compression is $2^{32}/32 = 134,217,728 \times$ at $z_0 \leq 1.2$ and $2^{32}/112 \approx 3.83 \times 10^7$ above it; overall

$$\frac{16 \times 2^{32}}{1,552} = \frac{2^{32}}{97} \approx 4.43 \times 10^7, \quad \text{against} \quad \frac{1,048,576}{496} \approx 2,114$$

The scaling is the point of the exercise. Passing from four parameters to five multiplies the raw lattice by $2^{16} = 65,536$ and the stored antichain by $1,552/496 \approx 3.13$: the object one must *not* store grows exponentially, the object one must store grows by a factor a little over three, and the compression ratio improves by roughly 2.1×10^4 . Exactness at five parameters is therefore not a constant

factor more expensive than at four; it is exponentially further out of reach of naive enumeration and essentially no further out of reach of the antichain.

0.9.4 Point-based evaluation at five parameters

The four-step α -set at $m = 5$ deduplicates $33^4 = 1,185,921$ action sequences to at most 15,929 distinct outcome vectors over the tested levels (11,145 at the top level), against $17^4 = 83,521 \rightarrow 545$ at $m = 4$ at the common top level: the raw enumeration grows $14.2\times$ and the top-level deduplicated object $20.4\times$, so the top-level compression ratio is *lower*, not higher, at five parameters. Across the 16 levels the four-cube maximum is 857 (at $z_0 = 2.2$), and the five-cube maximum is 15,929 (at $z_0 = 2.3$); neither top-level count is a global ceiling. Five-cube counts are 945 at $z_0 \leq 1.2$ and 11,145 at $z_0 = 2.5$, without monotonicity in between. The two endpoint behaviours of the four-cube persist: the one-step top-level collapse to a single outcome vector (33 sequences, one vector, every action safe in the silent region), and the edge behaviour (33 sequences, 33 distinct vectors, no collapse at all).

0.9.5 What is not claimed at six parameters and beyond

The ball construction gives $r^*(6) = 1$, so the 64 seven-cell radius-one balls of the six-cube are blind-survivable from every level by the same constant-matched-play argument. What is *not* determined at $m = 6$ is whether they are maximal: the distance- ≤ 3 graph there has 10,752 maximal cliques with maximum clique size 12, against the ball’s 7, and resolving their survivability needs a safety game in 12 dimensions, which the present apparatus does not settle. The five-cube classification above is complete; the six-cube is open, and the gap between 7 and 12 is where the next qualitative change, if there is one, would have to appear.

0.10 Related work, and what this paper is

This paper is a computational companion, and the combinatorics it leans on is classical. Both facts are stated here rather than left for a reader to discover.

0.10.1 The place of this paper in the series

The title’s “Scale II” refers to a companion relation, not to a volume number. The theory being computed here is the belief-state safety value $V_k(b)$ — the maximal probability of remaining safe for k steps under the disturbance’s worst case — together with its support identity and α -vector structure; that theory, and the Sperner-bounded antichain formula behind its witness census, is developed in the companion paper (Abaee, 2026, Belief-state safety values for viability under incomplete observation; in this set, Paper 2). The present paper takes that theory’s objects into the curse-of-dimensionality regime on one explicit instance and classifies the instance completely in exact arithmetic. It proves no new result about the theory itself. A reader wanting the theorems should read that paper; a reader wanting the instance classified, and the cost of doing so exactly, should read this one.

0.10.2 Antichains, Sperner theory and Dedekind’s problem

An antichain in the Boolean lattice is a family of pairwise incomparable subsets. Sperner’s theorem (Sperner, 1927) gives the maximum size of an antichain on an n -set as $\binom{n}{\lfloor n/2 \rfloor}$; counting *all* antichains is Dedekind’s problem, with the asymptotic count $2^{\binom{n}{\lfloor n/2 \rfloor}(1+o(1))}$ due to Kleitman (1966) and exact values known only for small n (Stanley, 2013; Engel, 1997, for the standard treatments). Hamming adjacency, and the coding-theoretic machinery built on it, is likewise classical — and the extremal role of Hamming balls on the cube is the subject of hypercube vertex isoperimetry: among all subsets of the cube of a given size, Hamming balls (the initial segments of the simplicial order) minimise the vertex boundary (Harper, 1966), with stability versions showing the extremal sets *are* Hamming balls.

That adjacency deserves one word of separation, because the present paper also finds Hamming balls extremal and reaches them by a different route. Here the balls arise *radially*: the constant matched action $u = \mathbf{1}$ gives a cell at Hamming distance k the constant drift $-\frac{1}{2} + \frac{m-2k}{5}$, which depends on the cell only through k , so the set on which it is nonnegative is a Hamming ball of radius $r^*(m) = \lfloor (2m - 5)/4 \rfloor$ automatically. Nothing is being minimised. The coincidence with the isoperimetric extremal sets is therefore structural rather than consequential: the survivable region is a ball because the drift is radial, not because balls optimise any boundary functional, and no result of this paper depends on Harper’s theorem.

This paper proves no new theorem about antichains. The census of Section 0.6 is a computation *on one instance*: 1,048,576 raw subset evaluations collapsing to 496 stored maximal sets. That figure is an output, and its value is that it was obtained exactly and in full. The classical results are the scaffold, not the contribution.

0.11 Antichain algorithms in formal verification

A separate literature uses antichains not as objects to be counted but as a compact *data structure*. When the sets an algorithm manipulates are downward closed under a preorder, storing only the maximal elements is canonical and can be exponentially more succinct than storing the set itself. That observation underlies the antichain algorithms introduced for checking universality and language inclusion of finite automata (De Wulf, Doyen, Henzinger and Raskin, 2006) and extended to Büchi automata and the automata-based approach to model checking (Doyen and Raskin), to LTL satisfiability and model checking (De Wulf, Doyen, Maquet and Raskin, 2008), and to LTL realizability and synthesis (Filiot, Jin and Raskin, 2009, 2011). The succinctness is a theorem rather than an empirical finding: De Wulf et al. exhibit families of automata on which the subset construction is exponential while the antichain algorithm is polynomial, and show that the basic operations stay polynomial in the size of the compact representation while that size may be exponentially smaller than the set represented. The same representation is used for games of imperfect information, where the winning belief states are downward closed and only the maximal el-

ements are stored.

This paper stores the antichain of maximal survivable subsets, so it uses the same representation, and **the compression itself should not be read as a new result**. Three things distinguish the uses. First, those algorithms compute *fixed points* — of a reachability, universality or realizability operator — whereas there is no fixed point here: the object is a complete classification of one finite instance by exhaustive exact enumeration. Second, those settings are *qualitative*: the question is whether a player wins, and the stored antichain is a winning region. Here the stored antichain is the identifying structure of a *quantitative* worst-case value function $V_k(b)$, a maximal probability of remaining safe; the stored sets are the supports of its α -vectors rather than a winning set. Third, the figures reported below (1,048,576 collapsing to 496; 68,719,476,736 to 1,552) are *measured costs of exhaustive exact classification at this size*, offered as cost figures; they are not a claimed asymptotic improvement on those algorithms, and nothing here bears on their complexity.

0.11.1 Point-based and exact computation for partially observed models

Point-based value iteration and its successors (Pineau, Gordon and Thrun, 2003; Shani, Pineau and Kaplow, 2013, for the survey) approximate belief-space value functions by propagating a finite set of belief points, with error bounds in the usual formulations. The distinction here is not the idea of a point set but the arithmetic: every quantity in this paper is an exact integer or rational, so the catalogue is not an approximation with a bound but a classification of the instance. That costs feasibility at scale — which is why the instance is a four-parameter cube and not a larger one, and why Section 0.15 says so. On the qualitative side, the complexity and decidability landscape for partially observed models under qualitative objectives is established (Chatterjee, Doyen and Henzinger, 2009); the companion paper’s position relative to that literature is stated there, and this paper adds nothing to it.

0.11.2 What is claimed

Given the above, the contribution is exactly five things.

1. **A complete classification of the four-parameter cube instance** — every maximal blind survivable set, every band, and the observation ladder, in exact integer and rational arithmetic, with no approximation and no sampling.
2. **The cost figures for doing so.** The census collapse (1,048,576 $\rightarrow$ 496), the four-step α -set deduplication (83,521 $\rightarrow$ 857 maximum, 545 at the top level), and the doubling of the observation ladder. These are the empirical content of the paper: they say what exactness costs at this size.
3. **The pairwise-Hamming structural restrictions as they apply here** — that every blind survivable set consists of Hamming-adjacent cells, and that no three cells of the four-cube are pairwise adjacent, so no blind policy of any period keeps three or more cells safe. The adjacency notion is classical; its consequence for this instance is the result.

4. **The complete classification of the five-parameter cube** (Section 0.9) — the enlargement that Section 0.15 previously left open. The maximal sets are the 32 radius-one balls at every level, plus one 80-set orbit of four-cell sets from $z_0 \geq 1.3$; the tetrahedral orbit meets the pairwise-Hamming bound with equality and is nevertheless not survivable, so that instrument is necessary and not sufficient from dimension five (Remark 12); and the census collapses 68,719,476,736 raw subsets to 1,552 stored sets, a compression 2.1×10^4 times better than at four parameters.

5. **Machine-checking of the layer**, described next.

0.12 Constrained and chance-constrained POMDPs

The objective computed here — the maximal probability of remaining safe for a horizon, $V_k(b)$ — is not peculiar to this series, and the literature that studies that objective should be named. Constrained Markov decision processes, in which a policy maximises an expected return subject to constraints on expected or probabilistic quantities, were systematised by Altman (1999). Under partial observability the same question gives constrained POMDPs, and the methods available there are approximate by construction: approximate linear programming over a belief-sampled basis (Poupart, Malhotra, Pei, Kim, Goh and Bowling, 2015), online forward search from the current belief (Undurti and How, 2010), and heuristic search over a belief-space AND/OR graph (Santana, Thiébaux and Williams, 2016). Chance-constrained dynamic programming carries the risk budget as an additional state variable and is likewise developed with continuous state spaces and discretisation in view (Ono, Pavone, Kuwata and Balaram, 2015).

Three distinctions keep the present computation separate, and they are distinctions of object rather than of accuracy. First, those formulations *optimise over policies* subject to a constraint; here $V_k(b)$ is the value *function*, computed as an object of study at every belief in a declared class, and the policy is a byproduct of the recursion rather than the object searched for. Second, the constraint there is a threshold a policy must meet — a feasibility question, answered by the existence or non-existence of a feasible policy — whereas what is tabulated here is the probability itself, as a function of the horizon, the prior weight and the initial state, whose *shape* is the result: the piecewise-linear α -vector support, and the antichain structure of Section 0.6. Third, every method named above is approximate — the linear programming relaxation gives bounds, point-based methods give values at probed beliefs with bounds elsewhere, and heuristic search is admissible but incomplete. Nothing in this paper is approximate: the values are exact rationals, and the classification of the five-cube in Section 0.9 is exhaustive rather than sampled.

The overlap should be stated plainly rather than minimised. The question “what is the greatest probability of remaining safe under partial observation?” is asked, and answered approximately, in that literature, so the *question* is not new and this paper does not claim it is. What is contributed is the exact answer on a declared finite instance, together with the structure of the value function that makes an exact answer representable at all.

0.13 Mechanized verification

Selected combinatorial components, not the whole manuscript, have Lean declarations. Sixteen modules of the project's Lean 4 formalization are devoted to this paper's objects, in namespace `Formalizations.EBC`: `EBC_Bands`, `EBC_Classification`, `EBC_Deadline`, `EBC_Dynamics`, `EBC_ExactBelief`, `EBC_Hamming`, `EBC_Ladder` and `EBC_Pairs`, with successive versions of several. A full comment-stripped header count of these 16 modules is 138 theorem declarations (zero lemma headers). The historical 134 counted only lines opening with `theorem`: it omitted four `@[simp]` `theorem` declarations, `EBC_Deadline.holdRep_zero`, `EBC_Deadline.matchedRep_zero`, `EBC_Dynamics.natToK_zero`, and `EBC_Dynamics.natToK_succ`. The prior provisional 135 report was also an incomplete static count; the reproducible source census supersedes it. Declaration-level coverage is narrower than section names: `pair_sum_bound` and `cor_hamming_m4` constrain four-cube pairs; `floor_pair_fails`, `pair_survives`, and `survivor_is_singleton_or_adjacent_pair` cover stated floor, threshold and finite-horizon conditions; `ladder_doubling` is a conditional mass-ratio identity; `deadline_law` proves the specified four-coordinate hold-then-matched schedule. No EBC declaration enumerates the 16-level census, the 84 alpha comparisons or the five-cube orbit.

No admitted gaps. The pinned default Lake target builds cleanly under Lean 4 v4.34.1 (60 of 60 jobs). In Lean an unfinished proof surfaces as the axiom `sorryAx` in the axiom footprint of every declaration depending on it; a source-level scan with comments stripped finds no `sorry`, `admit` or `axiom` declaration anywhere in the formalization layer, all sixteen EBC modules included. The EBC declarations checked in this source family use only those standard classical axioms or none; this does not erase the separate P3 native-evaluation dependencies disclosed below. **Current Lean provenance.** The committed project pin v4.34.1 was rebuilt in GitHub Actions (default Lake target: 60/60 jobs, exit 0; <https://github.com/MIKEAA2020/general-sustainability/actions/runs/36946730855>). The exact project source and two footprint-check `.lean` files are archived with SHA-256 manifests. A comment-aware scan of all 62 source files finds no `sorry`, `admit`, explicit `axiom` or `constant` declaration. Twenty-one named footprints contain no `sorryAx`; the aggregate-import environment exposes three source-level `native_decide` uses in the P3 support/value examples, yielding nine distinct generated axiom names under this pin. These are executable computational certificates that extend kernel-only trust to the Lean compiler/runtime; they are not admitted gaps and must not be described as zero axioms or as kernel-only proofs. Three preserved historical modules outside the aggregate import closure are not certified by the default build. The v4.14.0 comparison of the *current* source fails in the prelude and does not reproduce the old source run.

Scope. What is machine-checked is the discrete combinatorial core — the structure the enumeration walks. The enumeration itself, including the census of Section 0.6, is

a computation performed by the deposited scripts and is *not* formalized.

0.14 Verification methods

The deposited verification script checks the four-cube core in exact integer and rational arithmetic, standard library only. The five-cube clique census and cross-level alpha-mask maxima require separate exact enumeration; the clique witness uses `NetworkX` for graph enumeration, and `alpha_check.py` uses `NumPy` only to pack and deduplicate integer outcome masks. Their source-specific results accompany this alignment draft and are not covered by a green legacy script. The battery opens with the pair-sum bound over all 120 pairs and 17 actions; the adjacency-triangle check over all 560 triples; the exhaustive one-, two-, three-periodic classification with 60-step certificates; the ten ladder values by survivable-subset enumeration within each support; the 84 point-based pairings against definition-level enumeration; the census counts; the deadline thresholds by direct simulation; the dimension-scope battery (the pair-sum bound at $m = 5$ over all 496 pairs and 33 actions; the ball radii $r^*(m)$ for $m = 3, \dots, 8$ with the drift sign pattern and 60-step simulations of the radius-one ball at $m = 5$ and the 29-cell radius-two ball at $m = 7$); and selected headline strings in the original source. A passing string check does not recompute the five-cube alpha masks or prove the classification.

A second, independent apparatus. The script above enumerates the four-cube core; it does not prove the whole paper or independently cover every five-cube assertion. Alongside it, the results of The formal layer proves particular ordered-field combinatorial lemmas, not every statement in Sections 0.3–0.8. In particular `ladder_doubling` assumes an already-proved probability ratio and `deadline_law` treats the declared four-coordinate hold-then-matched sequence. Exact grid counts and point-based comparisons rest on separate finite arithmetic, not on the formal source. Agreement where hypotheses coincide is a cross-check, not a substitute for either apparatus.

What each result now rests on, stated plainly:

- **Formalized components.** The declared four-cube pair-sum, far-pair exclusion, floor-pair impossibility, and adjacent-pair construction have Lean declarations with their stated dimension, stock and finite-horizon hypotheses. `ladder_doubling` proves a conditional ratio, not the ten grid entries; `deadline_law` proves the specified hold-then-matched schedule for four coordinates and symbolic natural deadline. These do not formalize the five-cube or the entire band, ladder and census text.
- **Enumerated, not proved.** The census, the 84 point-based pairings, the $83,521 \rightarrow 857$ maximum across levels, and the dimension-scope battery at $m \neq 4$. These are finite checks over the cube; their content is the count, and a count is what the script supplies.
- **Simulated.** The 60-step ball certificates at $m = 5$ and $m = 7$, and the grid certification of the deadline thresholds. The five-cube orbit verdicts above instead

use a whole-horizon aggregate-loss argument and a two-action witness. For the deadline the simulation is now redundant — the threshold is proved — and is retained as an independent cross-check.

Where this text previously attributed a result to simulation or enumeration alone, the attribution above governs.

0.15 Delimitations

The blind class is declared and load-bearing: the policy may not depend on the belief, so a single action sequence governs every parameter branch and the pair-sum argument applies; a z -feedback policy could branch its actions across branches (the branches’ stock paths diverge) and is outside the classification, as in the class discipline of the companion theory (Abaee, 2026, An obstruction calculus), where the blind and observation-dependent recursions are shown to differ — the feedback family contains the blind one, strictly, under hypotheses of the same shape as this instance’s. That result is proved *in the companion’s model*; the correspondence with the cube here is one of form, not of type, so it motivates restricting to the blind class rather than proving that restriction necessary. It does, however, fix the meaning of the open question recorded above: since the two disciplines differ strictly somewhere, whether a non-blind discipline beats the pairs *on the cube* cannot be settled by transferring the companion’s strictness, and is left open. Drifts are deterministic per branch; probes are exact and cost no stock beyond the blind decline they interrupt. The pairs-only classification is scoped to $m = 4$: in dimension five the constant matched action already keeps the six-cell radius-one ball alive from every level (Remark 5), and the maximal sets for $m \geq 5$ are classified in Section 0.9 at $m = 5$, and left open at $m \geq 6$. Whether some non-blind discipline beats the pairs on the cube is not claimed either way. Scaling beyond thirty-two cells — where even the antichain’s input enumeration becomes the bottleneck — is the next step. Section 0.9 shows that the ball construction alone does not settle it: at $m = 6$ the balls have seven cells while the maximal cliques of the pairwise-necessary graph reach twelve, and closing that gap needs a safety game the present apparatus does not carry.

0.16 Conclusion

The exact discipline survives the curse of dimensionality and the cube repays it with structure a pencil would never find: Hamming governance of survivable sets, a doubling observation ladder, a silent region at the top of the stock grid, and a four-order antichain census. The stored object is the antichain; the argued instrument is the pairwise bound; and the two together are what make 1,048,576 subset evaluations collapse to 496 sets without loss. At five parameters the same two instruments, with the safety game in place of an averaging argument, carry 68,719,476,736 evaluations onto 1,552 sets — and they carry the sharper news that the pairwise bound has stopped being sufficient.

References

- Abaee, A. (2026).. An obstruction calculus for viability under incomplete observation. Submitted.
- Abaee, A. (2026). Computational viability certification under incomplete observation: a continuous-to-finite bridge with exact certificates, a solver-assisted certified campaign, and a finite-side reference library. Submitted.
- Abaee, A., 2026. Minimax dual certificates for the common-action obstruction. Manuscript submitted for publication.
- Abaee, A. (2026).. Probabilistic sufficiency for the obstruction calculus: belief-state safety values under partial observation. Submitted.
- Abaee, A., 2026. Robust viability of the 2J3KL limit reference point under a surplus-production map: policy scoring, expansion, and when catch cannot help. Zenodo. <https://doi.org/10.5281/zenodo.22552060>
- Abaee, A., 2026. An obstruction calculus for viability under incomplete observation (submitted).
- Althoff, M., et al., 2020. ARCH-COMP20 category report: continuous and hybrid systems with linear continuous dynamics. In: *ARCH20, 7th International Workshop on Applied Verification of Continuous and Hybrid Systems*, EPiC Series in Computing, vol. 74, pp. 16–48.
- Aubin, J.-P., 1991. *Viability Theory*.
- Altman, E., 1999. *Constrained Markov Decision Processes*. Chapman & Hall/CRC. CRC reprint 2021, doi:10.1201/9781315140223.
- Baccelli, F., Cohen, G., Olsder, G.J. and Quadrat, J.-P., 1992. *Synchronization and Linearity: An Algebra for Discrete Event Systems*.
- De Wulf, M., Doyen, L., Henzinger, T.A. and Raskin, J.-F., 2006. Antichains: a new algorithm for checking universality of finite automata. In: *Computer Aided Verification (CAV 2006)*, Lecture Notes in Computer Science, vol. 4144, pp. 17–30. Springer.
- De Wulf, M., Doyen, L., Maquet, N. and Raskin, J.-F., 2008. Antichains: alternative algorithms for LTL satisfiability and model-checking. In: *Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2008)*, Lecture Notes in Computer Science, vol. 4963, pp. 63–77. Springer.
- Doyen, L. and Raskin, J.-F., 2009. Antichains for the automata-based approach to model-checking. *Logical Methods in Computer Science*, 5(1).
- Chatterjee, K., Doyen, L., Henzinger, T.A., 2009. Qualitative analysis of partially-observable Markov decision processes. In: *Int. Symp. Math. Found. Comput. Sci. (MFCS 2009)*. LNCS, vol. 5734, pp. 635–646.
- Engel, K., 1997. *Sperner Theory*. Cambridge University Press.
- Farkas, J., 1902. Theorie der einfachen Ungleichungen. *Journal für die Reine und Angewandte Mathematik* **124**, 1–27.
- Filiot, E., Jin, N. and Raskin, J.-F., 2009. An antichain algorithm for LTL realizability. In: *Computer Aided Verification (CAV 2009)*, Lecture Notes in Computer Science, vol. 5643, pp. 263–277. Springer.
- Filiot, E., Jin, N. and Raskin, J.-F., 2011. Antichains and compositional algorithms for LTL synthesis. *Formal Methods in System Design*, 39, 261–296.
- Geretti, L., Dit Sandretto, J.A., Althoff, M., et al., 2020.

ARCH-COMP20 category report: continuous and hybrid systems with nonlinear dynamics. In: *ARCH20, 7th International Workshop on Applied Verification of Continuous and Hybrid Systems*, EPiC Series in Computing, vol. 74, pp. 49–75.

Harper, L.H., 1966. Optimal numberings and isoperimetric problems on graphs. *Journal of Combinatorial Theory*, 1, 385–393.

Helly, E., 1923. Über Mengen konvexer Körper mit gemeinschaftlichen Punkten. *Jahresbericht der Deutschen Mathematiker-Vereinigung* **32**, 175–176.

Kleitman, D.J., 1969. On Dedekind’s problem: the number of monotone Boolean functions. *Proc. Amer. Math. Soc.* **21**, 677–682.

Lovejoy, W.S., 1991. Computationally feasible bounds for partially observed Markov decision processes. *Operations Research* **39**, 162–175.

Minato, S. (1993). Zero-suppressed BDDs for set manipulation in combinatorial problems. In: *Proceedings of the 30th Design Automation Conference (DAC ’93)*, 272–277.

Ono, M., Pavone, M., Kuwata, Y. and Balaram, J., 2015. Chance-constrained dynamic programming with application to risk-aware robotic space exploration. *Autonomous Robots*, **39**, 555–571. doi:10.1007/s10514-015-9467-7.

Pineau, J., Gordon, G., Thrun, S., 2003. Point-based value iteration: an anytime algorithm for POMDPs. *Proc. IJCAI 2003*.

Poupart, P., Malhotra, A., Pei, P., Kim, K.-E., Goh, B. and Bowling, M., 2015. Approximate linear programming for constrained partially observable Markov decision processes. *Proceedings of the AAAI Conference on Artificial Intelligence*, **29**(1). doi:10.1609/aaai.v29i1.9655.

Santana, P., Thiébaux, S. and Williams, B., 2016. RAO*: an algorithm for chance-constrained POMDP’s. *Proceedings of the AAAI Conference on Artificial Intelligence*, **30**(1). doi:10.1609/aaai.v30i1.10423.

Shani, G., Pineau, J., Kaplow, R., 2013. A survey of point-based POMDP solvers. *Autonomous Agents and Multi-Agent Systems* **27**(1), 1–51.

Sperner, E., 1927. Ein Satz über Untermengen einer endlichen Menge. *Math. Z.* **27**, 544–548.

Stanley, R.P., 2013. *Topics in Algebraic Combinatorics*. Chapter 4, The Sperner property. <https://math.mit.edu/~rstan/algcomb/algcomb.pdf>.

Undurti, A. and How, J.P., 2010. An online algorithm for constrained POMDPs. In: *2010 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3966–3973. doi:10.1109/robot.2010.5509743.

Declarations

Funding

None.

Competing interests

None.

Data availability

No external data were used.

Code availability

The verification script `paper2_exact_belief_computation_v13_verification.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`). The source version reports that it reproduces the four-parameter bounds, bands, ladder values, pairings, census counts, and deadline thresholds. The five-parameter classification added after that version requires a separate verification record; coverage by this older script is not claimed.

Author contributions

A.A. conceptualized the entire work, wrote, reviewed and edited the manuscript.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.